Rashawn Grant-Rhooms

Brampton, ON | 647-514-1040 | grantrhr@mcmaster.ca | LinkedIn | GitHub

Education

Honours Bachelor of Science, Mathematics & Statistics — Aug. 2024 – Present
McMaster University — *Hamilton, ON*

- Minor in Business and Computer Science
- Relevant coursework: Calculus I–III, Linear Algebra, Probability and Statistics, Discrete Mathematics, Introduction to Programming, Neural Networks and Machine Learning, Mathematical Finance

Experience

Returns & Inventory Data Analyst — May 2025 – Aug. 2025
Bosch — *Mississauga, ON*

- Analyzed high-volume operational datasets by reconciling physical inventory with internal ERP records to identify process inefficiencies and automation opportunities
- Performed structured data validation and classification on SKU-level return data to maintain high accuracy across internal reporting workflows
- Processed and verified 100+ returned products per shift using shipment records, SKU identifiers, and internal inventory database systems
- Maintained 99%+ classification accuracy while supporting reporting processes for return trends across major retail partners including Lowe's and Home Depot
- Generated structured operational reports supporting data-driven decision making and internal process improvements

Projects

LookFirst – Driving Awareness Evaluation System — Sep. 2024 – Present
Python, MediaPipe, OpenCV, PostgreSQL, FastAPI, Pygame

- Developed a computer vision system evaluating driver situational awareness using head pose estimation, gaze behavior, and scene context across video, image, and panoramic inputs
- Extracted 3D facial orientation signals (*yaw, pitch, roll*) and transformed facial landmarks into structured behavioral features for downstream analytics and queryable datasets
- Designed backend API services using FastAPI to collect and store session telemetry, gaze events, and behavioral metrics in PostgreSQL
- Implemented temporal detection algorithms to quantify scan frequency, mirror checks, directional coverage, and reaction time across simulated driving scenarios

MNIST Handwritten Digit Classifier — Jan. 2025 – Feb. 2025
Python, NumPy

- Implemented a multiclass perceptron classifier from scratch in Python to recognize handwritten digits (0–9) from the MNIST dataset without using machine learning frameworks
- Developed a mini-batch gradient descent training procedure with configurable batch sizes to iteratively optimize model weights
- Trained and evaluated the classifier across the full dataset and selected optimal model parameters based on classification performance
- Analyzed model errors by generating histograms of misclassification frequency by digit class to identify systematic prediction weaknesses

Credit Risk Classification Model (Banking Dataset) — Jan. 2025 – Mar. 2025
Python, NumPy, pandas

- Developed machine learning models to classify credit risk using structured financial datasets including account balances, credit history, and loan duration
- Implemented Perceptron and Softmax logistic regression algorithms using gradient descent optimization in Python
- Performed exploratory data analysis, feature engineering, and dataset preprocessing using pandas and NumPy
- Evaluated model performance using classification metrics, misclassification analysis, and convergence diagnostics

Technical Skills

Programming: Python, SQL, C/C++, R, JavaScript, Java
Backend & Data Systems: PostgreSQL, MySQL, FastAPI, REST APIs, Data Modeling
Tools: Jupyter Notebook, Excel, Google Sheets
Libraries: pandas, NumPy, Matplotlib, OpenCV, PyTorch, MediaPipe, TensorFlow, scikit-learn